

KELT-23Ab

R-2309

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_260627 · created 2026-08-17T23:43:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461218.8539 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.027
Transit depth (Rp/Rs)^2	2.72 +/- 0.9 [%]
Orbital Inclination (inc)	89.79 +/- 1.48
Airmass coefficient 1 (a1)	1.432 +/- 0.017
Airmass coefficient 2 (a2)	-0.183 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.71
Optimal Annulus	6.99
Transit Duration (day)	0.0575 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.027 vs 0.132 (deviation 1.22σ)
Duration fit vs published	0.0575 d vs 0.0954 d (deviation 17.23σ)
Mid-time O–C	-1.0 min
Depth SNR	6.1
Residual scatter	1.18 %

Light curve

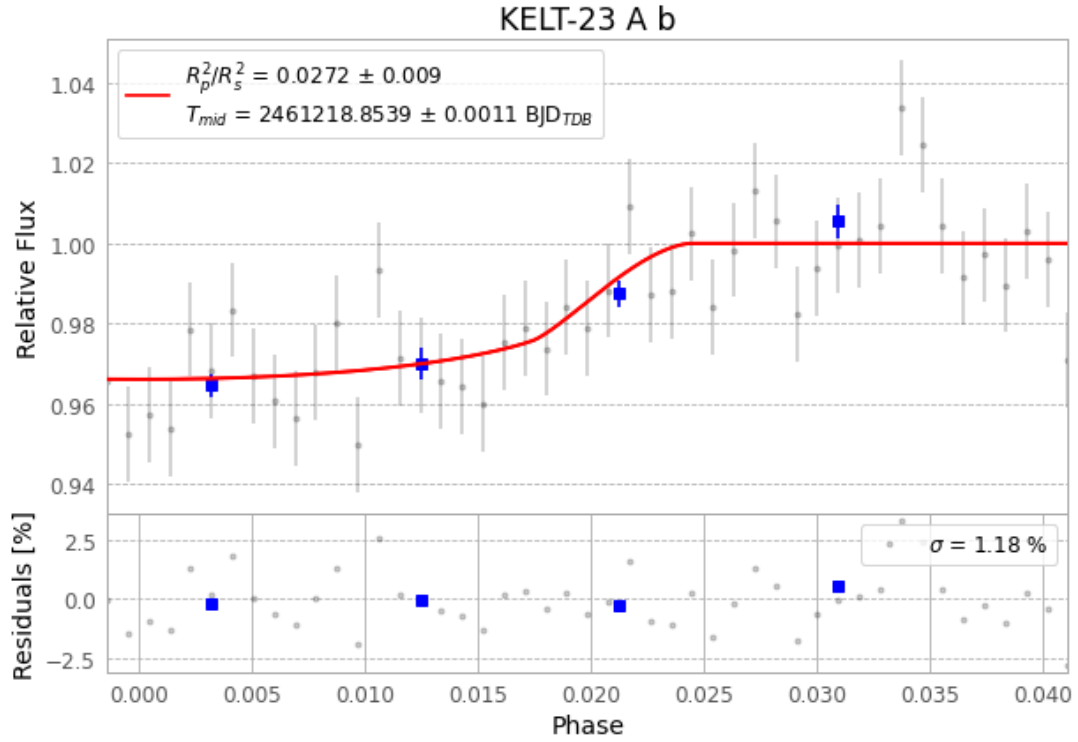


Figure 1 — FinalLightCurve_KELT-23 A b_2026-06-27.png (SHA-256 13275fe99db76a38...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 412a761ba8120eca...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

47 FITS frames (31 MB), KELT-23A260627082415.FITS → KELT-23A260627104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-260627.log (2c3a75fb8ded...), FinalLightCurve_KELT-23 A b_2026-06-27.png (13275fe99db7...), FinalLightCurve_KELT-23 A b_2026-06-27.csv (f95513c56a49...)

Compute resources

Wall time 97.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:43Z.